

RENAL PATHOLOGY

PRECLINICALS

@2025

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LEARNING OBJECTIVES

- **At the end of this lecture series, each student should be able to:**
- Review the anatomy and physiology of the kidneys
- Congenital diseases
- Glomerular diseases: primary and secondary,
- Tubulo-interstitial diseases
- Pyelonephritis
- Discuss the pathology of Cystic diseases of the kidney
- Tumours

INTRODUCTION

- The kidney is a structurally complex organ with various functions
 - ✓ Excrete harmful wastes
 - ✓ Maintains acid-base homeostasis
 - ✓ Absorbs essential substances
 - ✓ Regulates water and sodium metabolism
 - ✓ Maintains vascular tone
 - ✓ Produces erythropoietin
 - ✓ Maintains calcium homeostasis

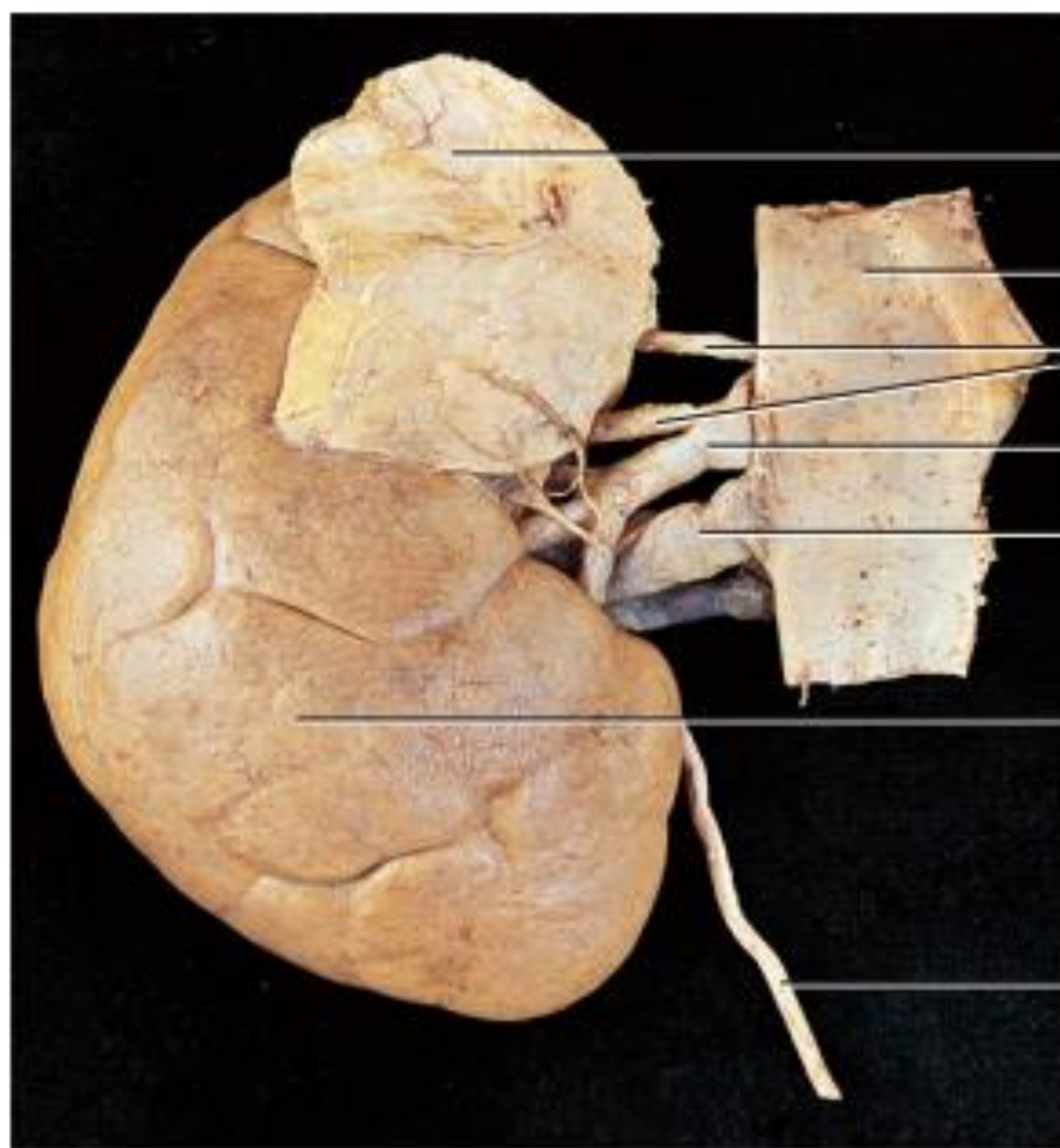
INTRODUCTION – CONT'D

- Diseases of the kidney are as complex as its structure but their study is facilitated by the splitting of the kidney into four components
- Early manifestations of diseases that affect each component tend to be distinctive
- Some components tend to be more vulnerable to specific forms of renal injury than others
- Whatever the origin, damage to all the four components of the kidney culminates in **end-stage kidney disease**

Kidney anatomy and physiology

- Are bean-shaped paired organs, each weighing about **150 gm** in the adult male and about **135 gm** in the adult female sitting in the retroperitoneum
- Performs many functions most importantly, filtration of unwanted substances
- **Renal cortex** forms the outer rim of the kidney and is about 1 cm in thickness
- contains all the glomeruli and about 85% of the nephron tubules
- **Renal medulla** is composed of 8-18 cone-shaped renal pyramids

LATERAL



MEDIAL

Adrenal (suprarenal) gland

Inferior vena cava

Suprarenal arteries

Renal artery

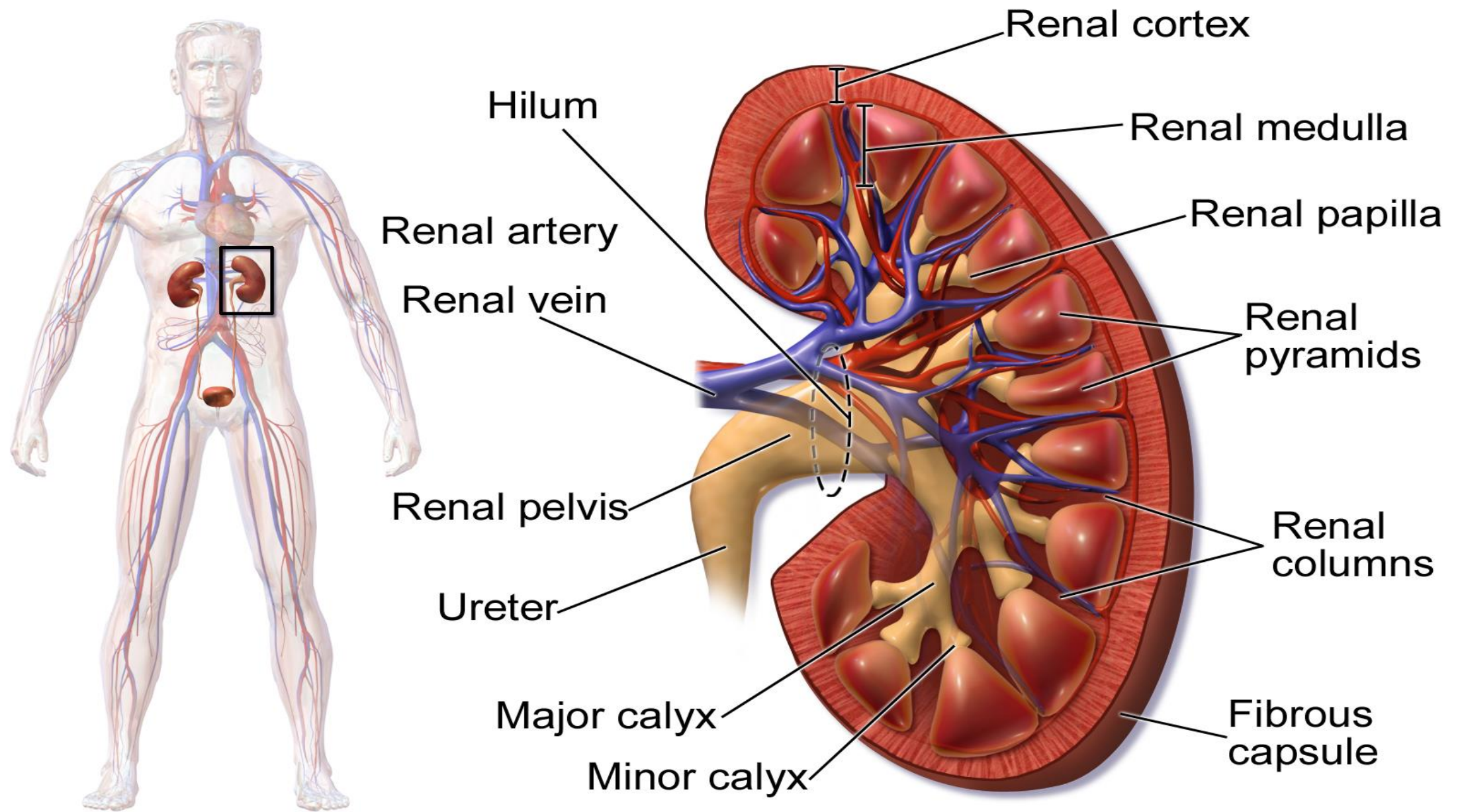
Renal vein

KIDNEY

Ureter

Kidney anatomy and physiology

- Remaining 15% nephrons consisting of collecting tubules, collecting ducts, loops of Henle and vasa recta send their loops into the medulla, and are therefore called **juxtamedullary nephrons**
- The base of a renal pyramid lies adjacent to the outer cortex and forms the **cortico-medullary junction**
- While the apex of each called the renal papilla contains the opening of each renal pyramid for passage of urine collected from collecting ducts and goes down into minor calyces.



Kidney Anatomy

SUPERIOR

Renal capsule

Renal cortex

Minor calyx

Major calyx

Renal artery

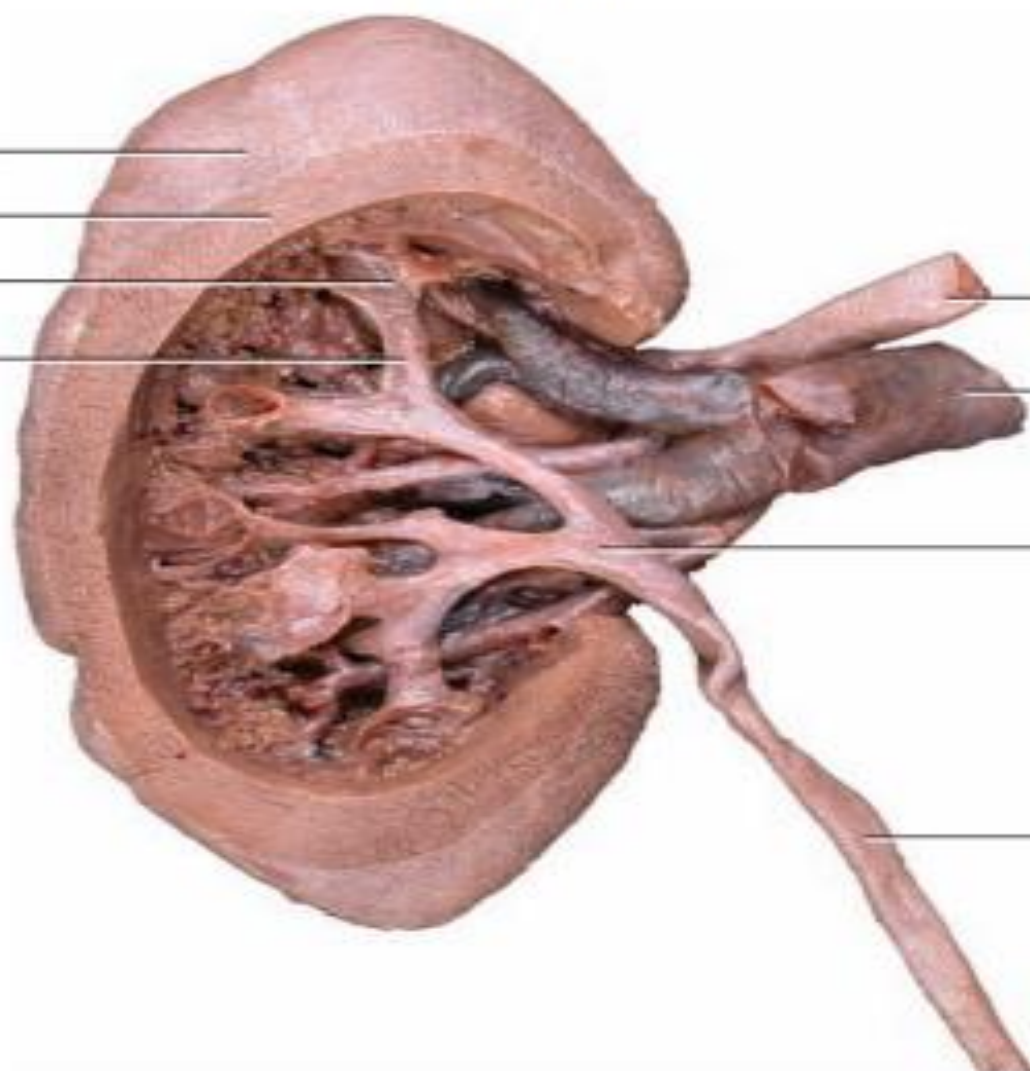
Renal vein

Renal pelvis

Ureter

LATERAL

MEDIAL



Kidney anatomy and physiology

- **Renal pelvis** is the funnel-shaped collection area of the urine for drainage into the ureter
- The minor calyces (8-18 in number in a normal kidney) collect urine from renal papillae and drain into major calyces (2-3 in a normal kidney)

Histology of the kidney

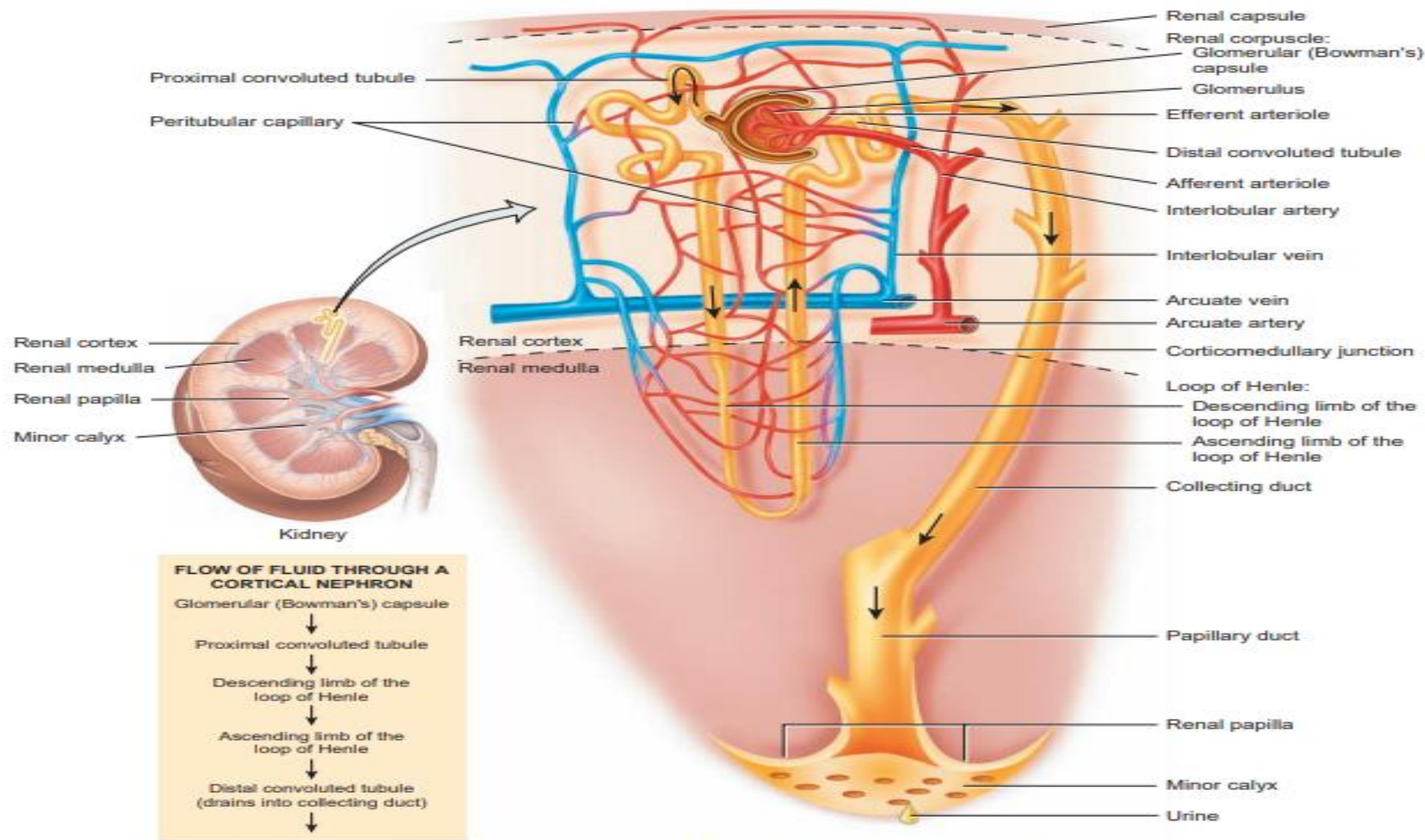
- Parenchyma of each kidney is composed of approximately one million microstructures called **nephrons**.
- A nephron, in turn, consists of 5 major parts, each having a functional role in the formation of urine:
 - the glomerular capsule (glomerulus and Bowman's capsule)
 - the proximal convoluted tubule (PCT)
 - the loop of Henle
 - the distal convoluted tubule (DCT) and
 - the collecting ducts

CONT'D

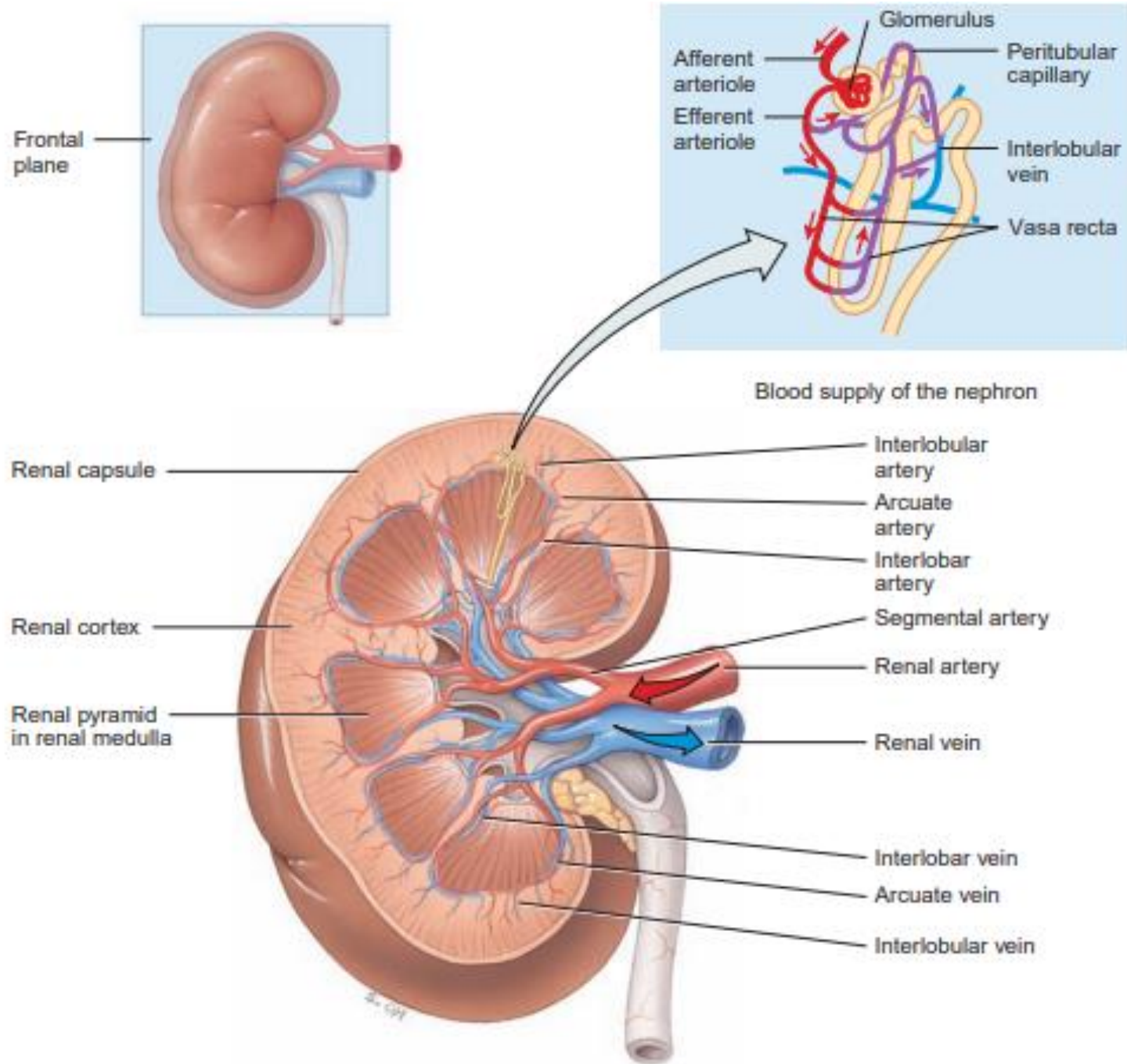
- From point of view of diseases of the kidneys, 4 components of renal parenchyma require further elaboration:
 - Renal vasculature
 - Glomeruli
 - Tubules
 - Interstitium

1. Renal vasculature

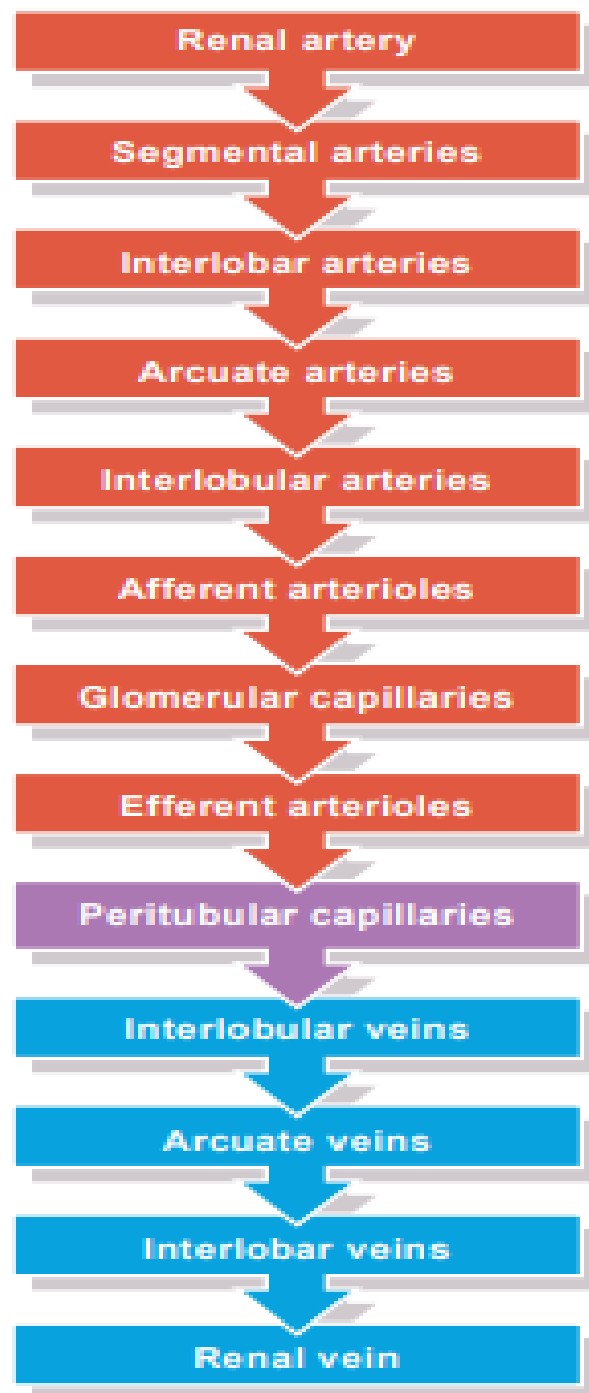
- Each kidney is supplied with blood by a **main renal artery** which arises from the aorta at the level of the 2nd lumbar vertebra → divides into anterior and posterior divisions → segmental branches from which interlobar arteries → give off the arcuate arteries which arch between the cortex and medulla → the afferent arterioles take their origin, each one supplying a single glomerulus → glomerulus emerge the efferent arterioles



(a) Cortical nephron and vascular supply



(a) Frontal section of right kidney



Renal vasculature – Cont'd

- The following important inferences can be drawn from the peculiarities of the renal vasculature:
- i) The renal cortex receives about 90% of the total renal blood supply and that the pressure in the glomerular capillaries is high.
- Therefore, renal cortex is **more prone** to the effects of **hypertension**
- ii) The renal medulla, on the other hand, is **poorly perfused** and any interference in blood supply to it results in **medullary necrosis**.

Renal vasculature – cont'd

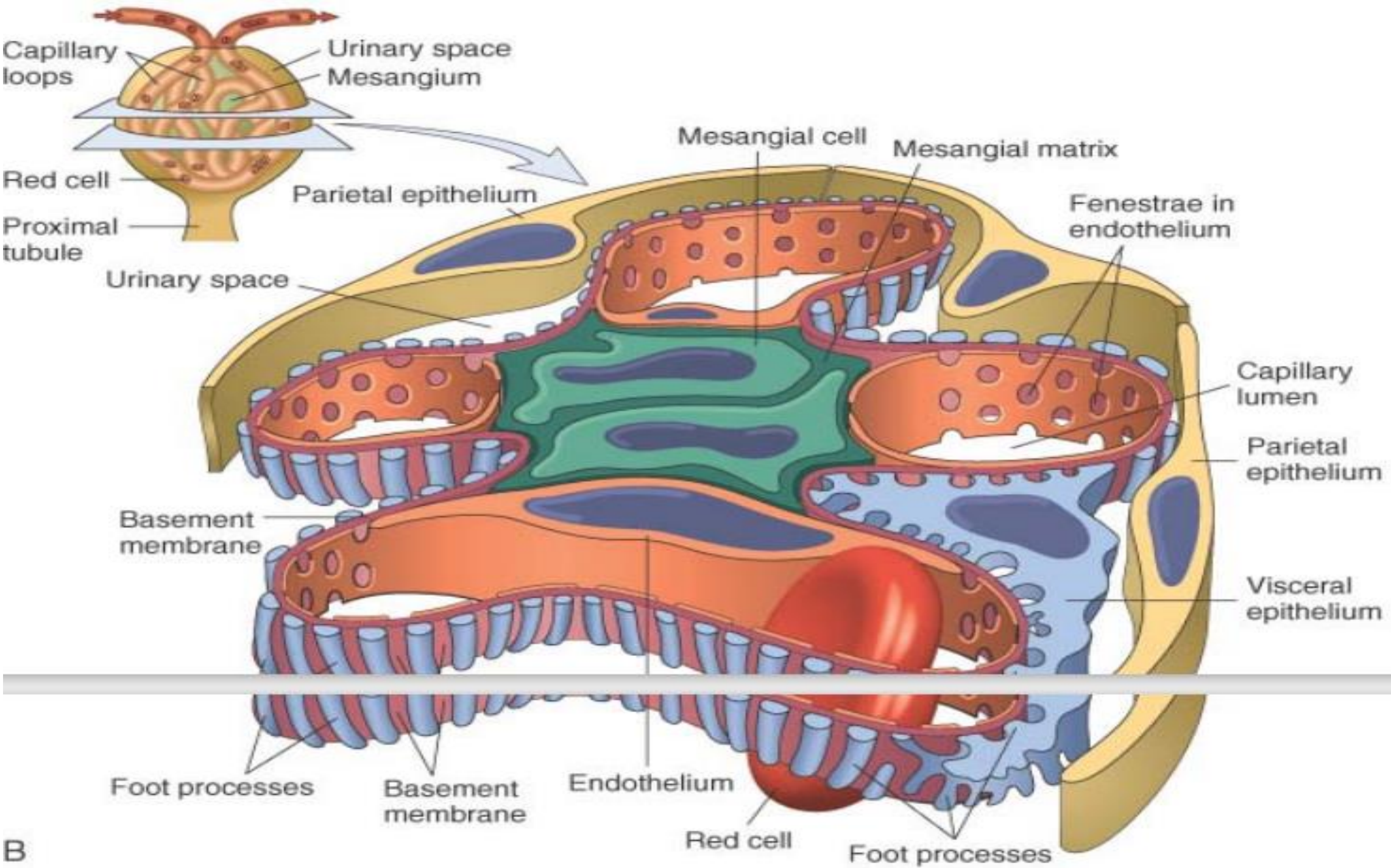
- iii) The divisions and subdivisions of the renal artery up to arterioles are end-arteries and have no anastomoses.
- Thus, **occlusion** of any of the branches results in infarction of the **renal parenchyma** supplied by it.
- iv) Since the tubular capillary beds are derived from the **efferent arterioles** leaving the glomeruli, diseases affecting the blood flow through glomerular tuft have significant effects on the tubules as well.

2. Glomerulus

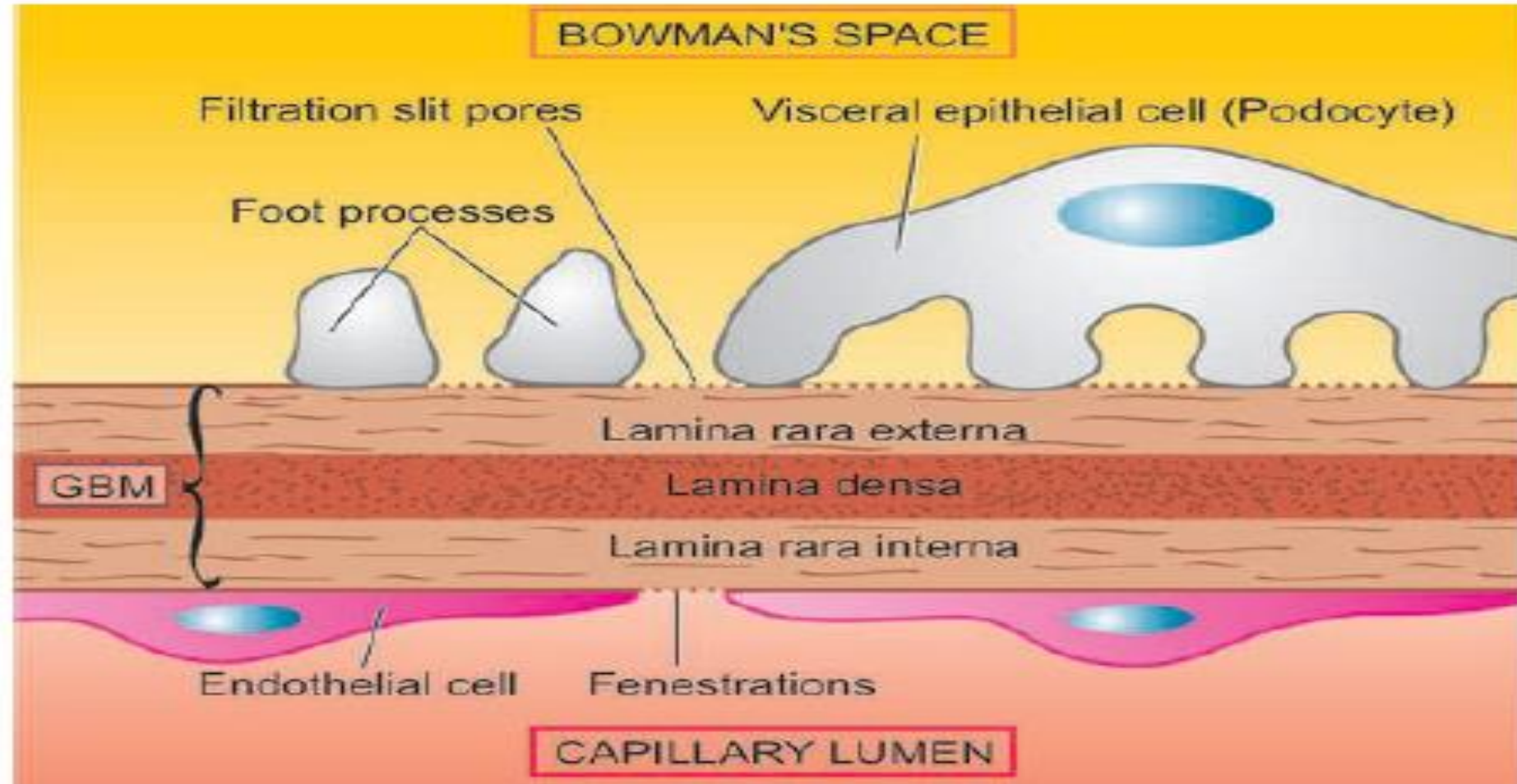
- The glomerulus consists of invagination of the blind end of the **proximal tubule** and contains a capillary tuft fed by the afferent arteriole and drained by efferent arteriole.
- The major function of glomerulus is **complex filtration** from the capillaries to the urinary space.
- Glomerular filtrate is quite similar in composition to plasma but lacks proteins and cells.
- Normally, **glomerular filtration rate** (GFR) is about **125 ml/minute**

Glomerulus – cont'd

- **Juxtaglomerular apparatus.** The juxtaglomerular apparatus (JGA) is situated at the vascular pole of the glomerulus
- The JGA is intimately concerned with sodium metabolism and is the principal source of **renin production**

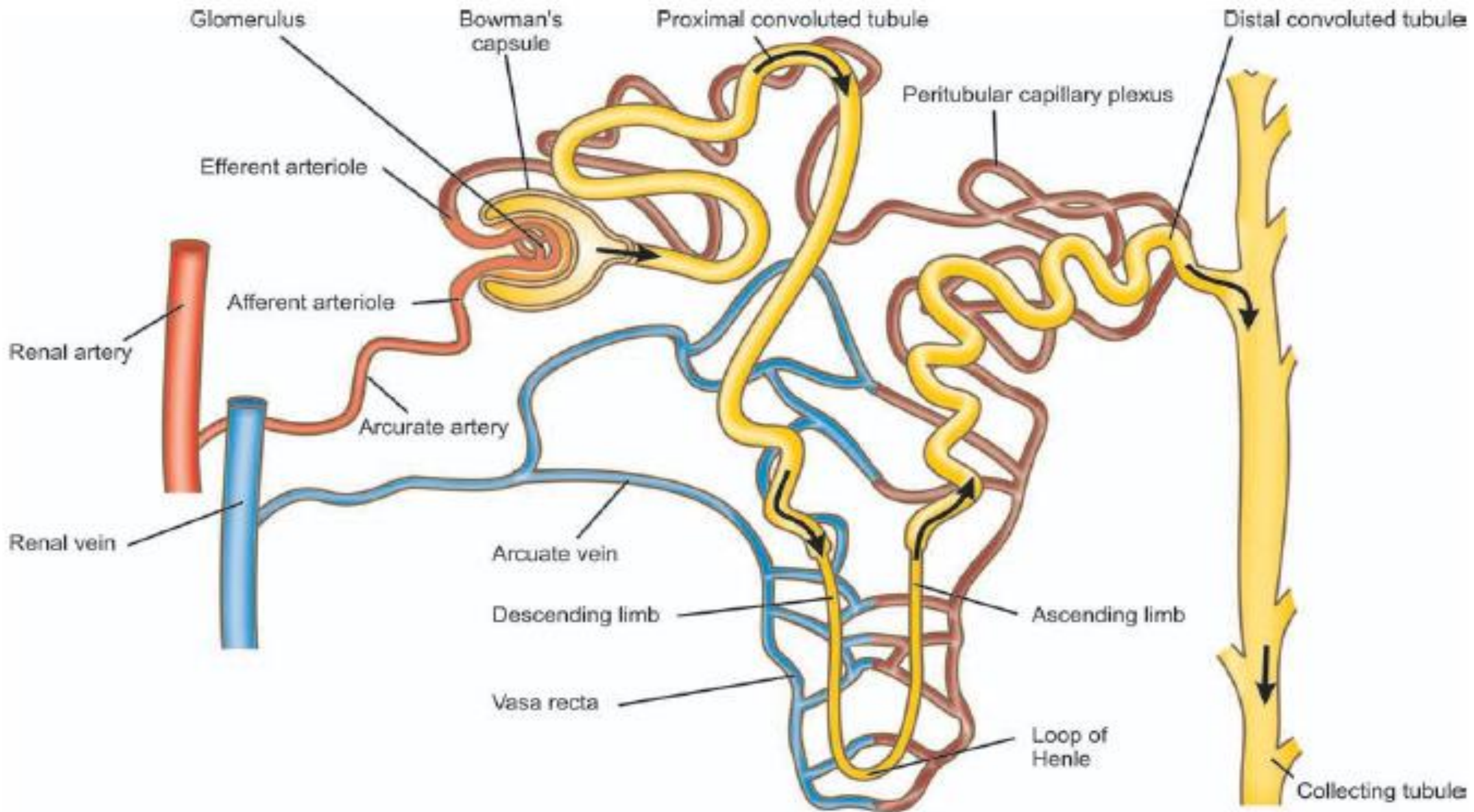


Ultrastructure of glomerular filtration barrier



3. Tubules

- The tubules of the kidney account for the greatest amount of the renal parenchyma.
- The structure of renal tubular epithelium varies in different parts of the nephron and is correlated with the functional capacity of that part of the tubule



Tubules – Cont'd

i) Proximal convoluted tubule (PCT)

- This is the first part arising from the glomerulus and is highly specialised part functionally.
- It is lined by cuboidal cells with a brush border composed of microvilli
- The major functions of PCT are: **active reabsorption** of filtered sodium, potassium, glucose, amino acids, proteins, vitamins, bicarbonate, phosphate, calcium and uric acid, and **passive reabsorption** of 80% of filtered water

Tubules – Cont'd

ii) Loop of Henle

- The PCT drains into the straight part of loop of Henle that consists of **thin descending**, and **thin and thick ascending limbs**, both of which have different structure and function.
- The descending limb is continuation of PCT, while ascending limb continues further into distal convoluted tubule (DCT).
- The descending segment of loop is lined by **simple epithelium** while the ascending limb is lined **by columnar cells**. The major function of loop of Henle is **active reabsorption** of sodium, potassium and chloride, and **passive diffusion** of water resulting in concentrated filtrate of urine.

Tubules – Cont'd

iii) Distal convoluted tubule (DCT)

- The DCT represents a transition from thick ascending limb from the point where the ascending limb meets the vascular pole of the glomerulus of its origin, to the early collecting ducts, lining cells in DCT are **cuboidal**.
- The epithelial cells at the point of beginning of DCT are taller, narrower and more closely packed to form the macula densa of JGA as already described.
- The DCT further contributes to **urinary concentration** and **acidification**, while the macula densa of JGA is the source of renin and has a role in sodium metabolism.

Tubules – Cont'd

iv) Collecting ducts

- The system of collecting ducts is the final pathway by which urine reaches the tip of renal papilla.
- The cells lining the collecting ducts are cuboidal but lack the brush border.
- Collecting ducts reabsorb water under control of **ADH**, and secrete **H⁺** and **K⁺ ions**.

4. Interstitium

- In health, the **renal cortical interstitium** is scanty and consists of a small number of fibroblast-like cells.
- But the medullary interstitium is more plentiful and contains stellate interstitial cells which are considered to produce an anti-hypertensive agent and are involved in the **metabolism of prostaglandins**.

Renal Failure

- Definition: this is a **condition** in which the kidneys fail to carry out its functions.

Renal Functions

- **Excretion** of water soluble waste products
- Maintenance of **extracellular fluid** (ECF) and composition – water and electrolytes
- Hormone **synthesis- renin, erythropoietin, and 1,25 di-hydroxy cholecalciferol**
- These are carried out by the functional unit of the kidney known as the **NEPHRON**

Pathophysiology of renal disease:

Renal failure

- Traditionally, diseases of the kidneys are divided into 4 major groups according to the predominant involvement of corresponding morphologic components:
- 1. **Glomerular diseases:** These are most often immunologically-mediated and may be acute or chronic.
- 2. **Tubular diseases:** These are more likely to be caused by toxic or infectious agents and are often acute.
- 3. **Interstitial diseases:** These are likewise commonly due to toxic or infectious agents and quite often involve interstitium as well as tubules (**tubulo-interstitial diseases**)

Cont'd

- 4. **Vascular diseases:** These include changes in the nephron as a consequence of increased intra-glomerular pressure such as in hypertension or impaired blood flow.
- The major morphologic involvements of the kidneys in the initial stage is confined to one component (glomeruli, tubules, interstitium or blood vessels), but eventually all components are affected leading to **end-stage kidneys**

Clinical Manifestations of Renal Diseases

- Regardless of cause, renal disease usually results in the evolution of one of the two major pathological syndromes:
- **Acute renal failure & Chronic renal failure**
- **Azotemia** is a biochemical abnormality that refers to an elevation of the **blood urea nitrogen** (BUN) and **creatinine levels**, and is related largely to a **decreased** glomerular filtration rate (GFR)
- Azotemia is a consequence of many renal disorders
- **Prerenal azotemia** is encountered when there is **hypoperfusion** of the kidneys (e.g., in hemorrhage, shock, volume depletion, and congestive heart failure) that impairs renal function in the absence of parenchymal damage.

Cont'd

- **Postrenal azotemia** is seen whenever urine flow is obstructed beyond the level of the kidney. Relief of the obstruction is followed by correction of the azotemia
- When azotemia becomes associated with a **constellation of clinical signs and symptoms** and **biochemical abnormalities**, it is termed **uremia**
- Uremia is characterized not only by failure of renal excretory function but also by a host of **metabolic** and **endocrine alterations** resulting from renal damage
- Uremic patients frequently manifest secondary involvement of the gastrointestinal system (e.g., uremic gastroenteritis), peripheral nerves (e.g., peripheral neuropathy), and heart (e.g., uremic fibrinous pericarditis)

Acute renal failure

- **Acute renal failure** (ARF) is a syndrome characterised by **rapid onset of renal dysfunction**, chiefly oliguria or anuria, and sudden increase in metabolic waste-products (urea and creatinine) in the blood with consequent development of uraemia.
- ETIOPATHOGENESIS. The causes of ARF may be classified as pre-renal, intra-renal and post-renal in nature
 - 1. **Pre-renal causes.** Pre-renal diseases are those which cause sudden decrease in blood flow to the nephron
- Renal ischaemia ultimately results in functional disorders or depression of GFR, or both
 - causes include
 - inadequate cardiac output and hypovolaemia or
 - vascular disease causing reduced perfusion of the kidneys

Cont'd

2. **Intra-renal causes.** Intra-renal disease is characterised by disease of renal tissue itself.

- include
 - vascular disease of the arteries and arterioles within the kidney,
 - diseases of glomeruli,
 - acute tubular necrosis due to ischaemia, or the
 - effect of a nephrotoxin,
 - acute tubulointerstitial nephritis and
 - pyelonephritis

Cont'd

3. **Post-renal causes.** Post-renal disease is characteristically caused by obstruction to the flow of urine anywhere along the renal tract distal to the opening of the collecting ducts

CLINICAL FEATURES. The clinical features will depend to a large extent on the underlying cause of ARF and on the stage of the disease at which the patient presents

- Oliguria or anuria
- Recent onset of azotemia
- Can result from glom, interstitial, vascular injury, acute tubular injury

Chronic Renal Failure (CRF)

- **Chronic renal failure** is a syndrome characterised by **progressive and irreversible deterioration** of renal function due to slow destruction of renal parenchyma, eventually terminating in death when sufficient number of nephrons have been damaged
- **Acidosis** is the major problem in CRF with development of biochemical azotaemia and clinical uraemia syndrome
- All chronic nephropathies can lead to CRF.
- The diseases leading to CRF can generally be classified into two major groups: those causing **glomerular pathology**, and those causing **tubulointerstitial pathology**

CRF – Cont'd

- Although exceptions abound, the evolution from normal renal function to symptomatic chronic renal failure broadly progresses through a series of four stages that merge into one another.
- 1. In **diminished renal reserve** the GFR is about **50%** of normal. Serum BUN and creatinine values are normal, and the patients are asymptomatic.
- However, they are more susceptible to developing azotemia with an additional renal insult.

CRF – Cont'd

- 2. In **renal insufficiency** the GFR is **20% to 50%** of normal. Azotemia appears, usually associated with anemia and hypertension.
- Polyuria and nocturia can occur as a result of decreased concentrating ability. Sudden stress (e.g., with nephrotoxins) may precipitate uremia
- 3. In chronic renal failure the GFR is less than **20% to 25%** of normal.
- The kidneys cannot regulate volume and solute composition, and patients develop edema, metabolic acidosis, and hyperkalemia.
- Overt uremia may ensue, with neurologic, gastrointestinal, and cardiovascular complications.

CRF – Cont'd

- 4. In **end-stage renal disease** the GFR is less than **5%** of normal; this is the terminal stage of uremia.
- Recent clinical classifications of chronic kidney disease, adopted in part to better stratify patients in clinical trials, adhere to this schema of progressive injury but divide patients into five classes based on levels of GFR

Clinical features of CRF

- Clinical manifestations of full blown CRF culminating in uraemic syndrome are described under 2 main headings: **primary (renal) uraemic manifestations** and **secondary (systemic or extra-renal) uraemic manifestations**
- A. **Primary uraemic (renal) manifestations.** Primary symptoms of uraemia develop when there is slow and progressive deterioration of renal function.

The resulting imbalances cause the following manifestations:

- Metabolic acidosis
- Hyperkalaemia
- Sodium and water imbalance
- Hyperuricaemia
- Azotaemia

Clinical features of CRF – Cont'd

- Secondary uraemic (extra-renal) manifestations. A number of extra-renal systemic manifestations develop secondarily following fluid-electrolyte and acid-base imbalances.
- These include the following:
 - Anaemia
 - Integumentary system
 - Cardiovascular system
 - Respiratory system
 - Digestive system
 - Skeletal system

Metabolic Alterations

- **1. Electrolyte Imbalance:** Uremia often results in disturbances in electrolyte levels, including hyperkalemia (high potassium), hyperphosphatemia (high phosphate), and hypocalcemia (low calcium). These imbalances can lead to muscle weakness, cardiac arrhythmias, and bone disease.
- **2. Acid-Base Disturbances:** The kidneys play a crucial role in maintaining acid-base balance. In uremia, metabolic acidosis can occur due to the accumulation of acidic waste products, leading to complications such as bone disease and muscle wasting.
- **3. Carbohydrate Metabolism:** Insulin resistance is common in uremia, contributing to glucose intolerance and an increased risk of diabetes. This resistance is partly due to the accumulation of uremic toxins that interfere with insulin signaling.

Cont'd

- **4. Lipid Metabolism:** Uremia can lead to dyslipidemia, characterized by elevated triglycerides and decreased high-density lipoprotein (HDL) cholesterol. This lipid profile contributes to the increased risk of cardiovascular disease in patients with chronic kidney disease.
- **5. Protein Metabolism:** Muscle wasting and protein-energy malnutrition are prevalent in uremic patients. This is due to a combination of factors, including inadequate dietary intake, metabolic acidosis, inflammation, and hormonal imbalances.

Endocrine Alterations

- **1. Parathyroid Hormone (PTH):** Secondary hyperparathyroidism is a common complication of chronic kidney disease. Elevated PTH levels result from decreased calcium absorption, phosphate retention, and decreased production of active vitamin D, leading to bone disease and vascular calcification.
- **2. Vitamin D Metabolism:** The kidneys are responsible for converting vitamin D to its active form, calcitriol. In uremia, decreased production of calcitriol contributes to bone disease and mineral metabolism disorders.
- **3. Erythropoietin (EPO):** The kidneys produce EPO, which stimulates red blood cell production. In uremia, decreased EPO production can lead to anemia, a common complication of chronic kidney disease.

Cont'd

- **4. Sex Hormones:** Uremia can disrupt the balance of sex hormones, leading to issues such as decreased libido, infertility, and menstrual irregularities.
- **5. Growth Hormone and Insulin-like Growth Factor-1 (IGF-1):** Growth retardation and malnutrition in children with chronic kidney disease can be attributed to alterations in the growth hormone/IGF-1 axis.

Glomerular Diseases

- Encompass a large and clinically significant group of renal diseases
- **Glomerulonephritis (GN)** or Bright's disease is the term used for diseases that primarily involve the renal glomeruli.
- It is convenient to classify glomerular diseases into 2 broad groups:
- I. **Primary glomerulonephritis** in which the glomeruli are the predominant site of involvement.
- II. **Secondary glomerular diseases** include certain systemic and hereditary diseases which secondarily affect the glomeruli
- The pattern & location of immune complex depositions are helpful in distinguishing among various types of GN

Clinicopathologic Classification of Glomerular Diseases

PRIMARY GLOMERULONEPHRITIS

- 1. Acute GN
 - i) Post-streptococcal
 - ii) Non-streptococcal
- 2. Rapidly progressive GN
- 3. Minimal change disease
- 4. Membranous GN
- 5. Membrano-proliferative GN
- 6. Focal proliferative GN
- 7. Focal segmental glomerulosclerosis (FSGS)
- 8. IgA nephropathy
- 9. Chronic glomerulonephritis

Clinicopathologic Classification of Glomerular Diseases – Cont'd

II. SECONDARY SYSTEMIC GLOMERULAR DISEASES

- 1. Lupus nephritis (SLE)
- 2. Diabetic nephropathy
- 3. Amyloidosis
- 4. Polyarteritis nodosa
- 5. Wegener's granulomatosis
- 6. Goodpasture's syndrome
- 7. Henoch-Schönlein purpura
- 8. Systemic infectious diseases (*bacterial* e.g. bacterial endocarditis, syphilis, leprosy; *viral* e.g. HBV, HCV, HIV; *parasitic* e.g. falciparum malaria, filariasis)
- 9. Idiopathic mixed cryoglobulinaemia

Clinical manifestations

- Is quite variable but in general four features—**proteinuria, haematuria, hypertension** and **disturbed excretory function**
- A firm diagnosis, however, can be established by examination of renal biopsy under light, electron and immunofluorescence microscopy
- A number of clinical syndromes are recognised in glomerular diseases.
- The following are major glomerular syndromes commonly found in different glomerular diseases:
 - **Nephritic and Nephrotic syndromes;**
 - **Asymptomatic proteinuria and Haematuria.**

Acute nephritic syndrome

- This is the **acute** onset of **haematuria**, **proteinuria**, **hypertension**, **oedema** and **oliguria** following an **infective illness** about 10 to 20 days earlier.
- 1. **haematuria** is generally slight giving the urine smoky appearance and erythrocytes are detectable by microscopy or by chemical testing for haemoglobin.
- Appearance of red cell casts is another classical feature of acute nephritic syndrome.
- 2. The **proteinuria** is mild (**>150mg** but less than **3 gm per 24 hrs**) and is usually non-selective (nephritic range proteinuria).
- 3. **Hypertension** is variable depending upon the severity of the glomerular disease but is generally mild.

ANS – CONT'D

- 4. **Oedema** in nephritic syndrome is **usually mild** and results from **sodium and water retention**
- 5. **Oliguria** is variable and reflects the severity of glomerular involvement
- Underlying causes of acute nephritic syndrome may be primary glomerulonephritic diseases (**classically acute glomerulonephritis** and **rapidly progressive glomerulonephritis**)

Causes of Acute Nephritic Syndrome

I. PRIMARY GLOMERULONEPHRITIS

1. Acute GN
 - i) Post-streptococcal
 - ii) Non-streptococcal
2. Rapidly progressive GN
3. Membranoproliferative GN
4. Focal GN
5. IgA nephropathy

II. SYSTEMIC DISEASES

1. SLE
 2. Polyarteritis nodosa
 3. Wegener's granulomatosis
 4. Henoch-Schonlein purpura
 5. Cryoglobulinaemia
-

Nephrotic Syndrome

- is a constellation of features in different diseases having varying pathogenesis; it is characterised by findings of **massive proteinuria**, **hypoalbuminaemia**, **oedema**, **hyperlipidaemia**, **lipiduria**, and **hypercoagulability**.
- 1. **Heavy proteinuria** (protein loss of **more than 3 gm per 24 hrs**) is the chief characteristic of nephrotic syndrome (nephrotic range proteinuria).
- 2. **Hypoalbuminaemia** is produced primarily consequent to urinary loss of albumin, and partly due to increased renal catabolism and inadequate hepatic synthesis of albumin.

Nephrotic Syndrome - Contd

- Often, the plasma albumin level is 1 to 3 gm/dl (normal 3.5 to 5.5 gm/dl) and there is reversed albumin-globulin ratio.
- 3. **Oedema** in nephrotic syndrome appears due to fall in colloid osmotic pressure consequent upon hypoalbuminaemia
- Sodium and water retention further contribute to oedema
- 4. **Hyperlipidaemia** is a frequent accompaniment of nephrotic syndrome. The exact mechanism of its genesis is not clear
- 5. **Lipiduria** occurs following hyperlipidaemia due to excessive leakiness of glomerular filtration barrier.

Nephrotic Syndrome - Contd

- 6. **Hypercoagulability**. Patients with nephrotic syndrome may develop spontaneous arterial or venous thrombosis, renal vein thrombosis and pulmonary embolism due to various factors:
 - increased urinary loss of antithrombin III,
 - hyperfibrinogenaemia from increased synthesis in the liver,
 - decreased fibrinolysis,
 - Increased platelet aggregation
 - altered levels of protein C and S

Causes of Nephrotic Syndrome

I. PRIMARY GLOMERULONEPHRITIS

1. Minimal change disease (*most common in children*)
2. Membranous GN (*most common in adults*)
3. Membranoproliferative GN
4. Focal segmental glomerulosclerosis
5. Focal GN
6. IgA nephropathy

II. SYSTEMIC DISEASES

1. Diabetes mellitus
2. Amyloidosis
3. SLE

III. SYSTEMIC INFECTIONS

1. Viral infections (HBV, HCV, HIV)
2. Bacterial infections (bacterial endocarditis, syphilis, leprosy)
3. Protozoa and parasites (*P. falciparum* malaria, filariasis)

IV. HYPERSENSITIVITY REACTIONS

1. Drugs (heavy metal compounds like gold and mercury, other drugs like penicillamine, trimethadione and tolbutamide, heroin addiction)
2. Bee stings, snake bite, poison ivy

V. MALIGNANCY

1. Carcinomas
2. Myeloma
3. Hodgkin's disease

VI. PREGNANCY

Toxaemia of pregnancy

VII. CIRCULATORY DISTURBANCES

1. Renal vein thrombosis
2. Constrictive pericarditis

VIII. HEREDITARY DISEASES

1. Alport's disease
 2. Fabry's disease
 3. Nail-patella syndrome
-

Contrasting Features of Acute Nephritic and Nephrotic Syndromes

Feature	Acute Nephritic Syndrome	Nephrotic Syndrome
1. <i>Proteinuria</i>	Mild (< 3 gm per 24 hrs)	Heavy (> 3 gm per 24 hrs)
2. <i>Hypoalbuminaemia</i>	Uncommon	Present
3. <i>Oedema</i>	Mild, in loose tissue	Marked, generalised peripheral
4. <i>Mechanism of Oedema</i>	Na ⁺ and water retention	↓ plasma osmotic pressure, Na ⁺ and water retention
5. <i>Haematuria</i>	Present, microscopic	Absent
6. <i>Hypertension</i>	Present	Present in advanced disease
7. <i>Hyperlipidaemia</i>	Absent	Present
8. <i>Lipiduria</i>	Absent	Present
9. <i>Oliguria</i>	Present	Present in advanced disease
10. <i>Hypercoagulability</i>	Absent	Present

Acute renal failure

- As already described, acute renal failure (ARF) is characterised by rapid decline in renal function.
- ARF has many causes including glomerular disease, principally rapidly progressive GN and acute diffuse proliferative GN

Chronic renal failure

- These cases have advanced renal impairment progressing over years and is detected by significant proteinuria, haematuria, hypertension and azotaemia
- Such patients generally have **small contracted kidneys** due to chronic glomerulonephritis

Asymptomatic proteinuria

- Presence of proteinuria unexpectedly in a patient may be unrelated to renal disease (e.g. exercise-induced, extreme lordosis and orthostatic proteinuria), or may indicate an underlying mild glomerulonephritis
- Association of asymptomatic haematuria, hypertension or impaired renal function with asymptomatic proteinuria should raise strong suspicion of underlying glomerulonephritis

Asymptomatic haematuria

- Asymptomatic microscopic haematuria is common in children and young adolescents and has many diverse causes such as diseases of the glomerulus, renal interstitium, calyceal system, ureter, etc
- GH is indicated by the presence of **red blood cells**, **red cell casts** and **haemoglobin** in the urine & frequently associated with proteinuria

Pathogenesis of glomerular injury (summary)

- Most forms of primary GN and many of the secondary glomerular diseases in human beings have immunologic pathogenesis
- The consequences of injury at different sites within the glomerulus in various glomerular diseases can be assessed when compared with the normal physiologic role of the main cells involved i.e. **endothelial**, **mesangial**, **visceral epithelial**, and **parietal epithelial** cells as well as of the **GBM**
- Immunologic mechanisms underlying glomerular injury are primarily antibody-mediated (immune-complex disease).

A. Antibody-Mediated Glomerular Injury

- 1. IMMUNE COMPLEX DISEASE. Majority of cases of glomerular disease result from deposits of immune complexes (**antigen-antibody complexes**)
- The immune complexes are represented by irregular or granular glomerular deposits of immunoglobulins (IgG, IgM and IgA) and complement (mainly C3).
- 2. ANTI-GBM DISEASE. Less than 5% cases of human GN are associated with **anti-GBM antibodies**
- The constituent of GBM acting as **antigen** appears to be a component of **collagen IV** of the basement membrane.

B. Cell-mediated Glomerular Injury (Delayed-type Hypersensitivity)

- Cytokines and other mediators released by activated T cells stimulate cytotoxicity, recruitment of more leucocytes and fibrogenesis.
- **CD4+** T lymphocytes recruit more macrophages while **CD8+** cytotoxic T lymphocytes and natural killer cells cause further glomerular cell injury by antibody-dependent cell toxicity
- However, cell-mediated injury is yet less clear than antibody-mediated glomerular injury.

PRIMARY GLOMERULONEPHRITIS

Acute proliferative (poststreptococcal, postinfectious) glomerulonephritis

- Most common underlying infections are streptococcal
- Occurs 1-4 weeks after a streptococcal infection of pharynx or skin
- Occurs mostly in children 6 – 10 yrs
- Only certain strains of group A β -hemolytic streptococci are nephritogenic
- Immunologically mediated disease
- $\uparrow$ titers of Abs against streptococcal Ags present - $\uparrow$ ASO
- Serum complement levels low – C3
- Granular immune deposits in glomeruli – immune complex-mediated mechanism

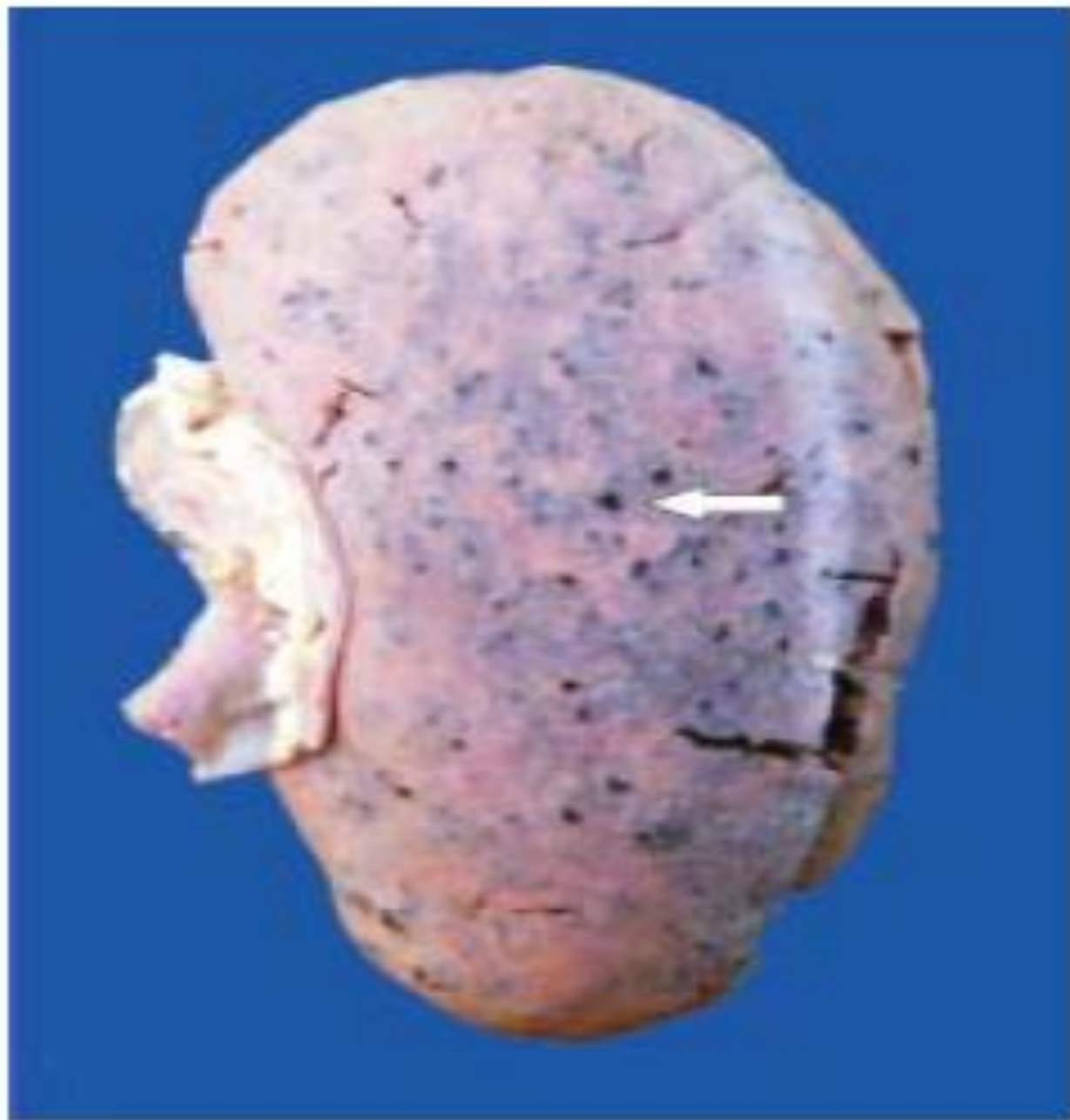
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Morphology

- Kidneys swollen and “flea-bitten”
- LM – Glom hypercellular due to proliferation of mesangial, endothelial, epithelial cells
- IF – granular deposition of IgG, C3 in subepithelial region

Clinical course

- 95% children recover within 3 years
- A few may progress to chronic GN or RPGN
- Prognosis poor in adults in the setting of hypertension



Rapidly progressive (crescentic) GN

- Clinicopathologic syndrome with rapid progression to ARF
- Patients present with severe oliguria and signs of nephritic syndrome
- Microscopically **crescents** are present in glom
- Caused by different diseases affecting kidney or systemic
- Glom injury is immunologically mediated
- Three types of RPGN
 - Type I – anti-GBM antibody
 - Type II – immune complex
 - Type III – pauci-immune

Cont'd

Morphology

- Kidneys enlarged and pale
- Formation of crescents – due to proliferation of parietal cells and presence of inflammatory cells

Clinical course

- Presents with nephritic syndrome, prognosis general dismal
- In type I RPGN (Goodpasture syndrome) hemoptysis and pulmonary hemorrhage may be prominent features
- Treatment includes plasmapheresis, steroids, cytotoxic agents

Membranous nephropathy

- Common cause of **nephrotic syndrome** in adults
- Characterised by diffuse thickening of glom capillary wall due to accumulation of deposits on GBM
- Secondary MG – occurs in association with systemic diseases – drugs, cancer, SLE, infections
- In 85% patients disease is idiopathic

Cont'd

Morphology

- Kidneys large, swollen and pale
- LM – diffuse thickening of capillary wall : MAC of the CS
- IF – GBM material accumulates between deposits forming “spikes”
- Loss of foot processes

Clinical course

- Disease generally indolent
- Does not respond well to corticosteroids
- Spontaneous remission may occur
- 40% may progress to chronic RF

Minimal-change disease

- Most common cause of nephrotic syndrome in children, less common in adults
- Diffuse effacement of foot processes in glom that appear normal by LM
- Peak 2 – 6 years of age
- Etiology remains elusive but majority cases are idiopathic, rest associated with systemic diseases
- May follow a respiratory tract infection or immunisation
- Dramatic response to steroids therapy

Cont'd

Morphology

- LM – glom normal, no deposits seen
- EM – GBM appear normal
 - Podocytes show diffuse effacement of foot processes

Clinical course

- Despite massive proteinuria renal function remains good
- Proteinuria is highly selective – albumin
- More than 90% children respond rapidly to steroids
- Long term prognosis excellent

Focal segmental glomerulosclerosis

- is a condition in which there is **sclerosis** and **hyalinosis** of some glomeruli & portions of their tuft (less than 50% in a tissue section),
- May occur as primary disease – idiopathic FSGS
- In association with other known conditions – HIV infection, heroin addition, SCD, obesity
- Idiopathic cases account for 35% of nephrotic syndrome in adults
- Proteinuria is nonselective
- Poor response to steroid therapy
- Up to 50% patients progress to end stage renal disease

Cont'd

Morphology

- Focal and segmental collapse of capillary structure
- EM – focal loss of epithelial cells
 - thickening of capillary basement membrane
- Protein enters mesangium, initiates a reaction → fibrosis
- Progression to global **sclerosis** with time
- **Hyalinosis** refers to collection of eosinophilic, homogeneous, PAS-positive, hyaline material present on the inner aspect of a sclerotic peripheral capillary loop

Membranoproliferative GN

- Also called mesangiocapillary GN
- Causes up to 20% nephrotic syndrome in children and young adults
- Can be primary (idiopathic) or secondary to other systemic disorders
- Primary MPGN divided into 2 types, type I and II (dense deposit disease)
- MPGN is characterised histologically by:
 - Alterations in glomerular basement membrane
 - Proliferation of glomerular cells
 - Leukocyte infiltration

Cont'd

Pathogenesis

- Immune complexes are formed in glom with activation of complement pathways
- Antigens are unknown
- Some are proteins from microbes e.g. HBV, HCV
- These are planted in glom or are trapped or are preformed immune complexes from circulation

Cont'd

Morphology

- LM – glom large and hypercellular
 - Hypercellularity due to proliferation of cells and infiltration by leukocytes
 - Crescents may be present
 - GBM is thickened , glom capillary wall shows “double contour” or “tram-track” appearance due to duplication of BM
 - Subendothelial electron-dense deposits
- In type II – deposition of dense material of unknown composition in GBM

Focal Proliferative Glomerulonephritis

- Frequently manifests as nephrotic syndrome characterised by pathologic changes in certain number of glomeruli (**focal**),
- And often confined to one or two lobules of the affected glomeruli (**segmental**), while other glomeruli are normal
- is, thus, a pathologic diagnosis
- It may occur under following diverse clinical settings:
 - early manifestation of a number of systemic diseases
 - component of a known renal disease such as in IgA nephropathy
 - a primary idiopathic glomerular disease unrelated to systemic or other renal disease

Cont'd

Morphology

- LM - Focal and segmental glomerular involvement
- focal and segmental cellular proliferation of mesangial cells and endothelial cells
- IM – widespread mesangial deposits of immunoglobulins (mainly IgA with or without IgG), complement (C3) and fibrin
- Clinical features
 - Haematuria
 - Proteinuria is frequently mild to moderate
 - hypertension is uncommon

Chronic Glomerulonephritis

- Is the final stage of a variety of glomerular diseases which result in irreversible impairment of renal function
- May progress to chronic GN, in descending order of frequency, are as under:
 - i) Rapidly progressive GN (90%)
 - ii) Membranous GN (50%)
 - iii) Membranoproliferative GN (50%)
 - iv) Focal segmental glomerulosclerosis (50%)
 - v) IgA nephropathy (40%)
 - vi) Acute post-streptococcal GN (1%)
- About **20% cases** of chronic GN are idiopathic without evidence of preceding GN of any type

Cont'd

Morphology

- kidneys are usually small and contracted weighing as low as 50 gm each
- the cortex is narrow and atrophic, while the medulla is unremarkable

Clinical Features

- Patients are usually adults
- Characterised by hypertension, uraemia and progressive deterioration of renal function
- There are a variety of systemic manifestations of uraemia
- Patients eventually die if they do not receive a renal transplant

Type	Clinical Features	Pathogenesis	Pathology		
			LM	EM	IFM
1. <i>Acute GN</i>	Acute nephrotic syndrome	Immune complex disease (local or circulating)	Diffuse proliferation, leucocytic infiltration	Subepithelial deposits ('humps')	Irregular IgG, C3
2. <i>RPGN</i>	Acute renal failure	i) Type I: anti-GBM type ii) Type II: immune complex type iii) Type III: pauci-immune RPGN	Proliferation, crescents	i) Linear deposits along GBM ii) Subepithelial deposits iii) No deposits	i) Linear IgG, C3 ii) Granular IgG, C3 iii) Negative
3. <i>Minimal change disease</i>	Nephrotic syndrome (highly selective proteinuria)	Reduction of normal negative charge on GBM ?Cell-mediated mechanism	Normal glomeruli, lipid vacuolation in tubules	Loss of foot processes, no deposits	Negative
4. <i>Membranous GN</i>	Nephrotic syndrome	Immune complex disease (local)	Diffuse thickening of capillary wall	Subepithelial deposits ('spikes')	Granular IgG, C3

5. <i>Membrano-proliferative GN</i>	Nephrotic syndrome	Type I: immune complex disease Type II: dense deposit disease (alternate pathway activation) Type III: rare, with systemic diseases and drugs	Lobular proliferation of mesangial cells, increased mesangial matrix, double contour of GBM	Type I: Subendothelial deposits Type II: Dense intramembranous deposits Type III: Subendothelial and subepithelial deposits	Type I: IgG, C3 Type II: C3 properdin Type III: C3 IgG, IgM
6. <i>Focal GN</i>	Variable, haematuria common	Variable, possibly immune complex disease	Focal and segmental proliferation	Mesangial deposits	IgA ± IgG, C3 and fibrin
7. <i>Focal Segmental glomerulosclerosis</i>	Nephrotic syndrome	i) Idiopathic ii) With superimposed primary glomerular disease iii) Secondary type	Focal and segmental sclerosis and hyalinosis	Loss of foot processes, electron dense deposits in regions of sclerosis and hyalinosis	IgM, C3

8. <i>IgA nephropathy</i>	Recurrent haematuria, mild proteinuria	Unknown, possibly alternate pathway disease	Variable, commonly focal proliferative GN	Mesangial electron-dense deposits	IgA $\pm$ IgG, C3, properdin
9. <i>Chronic GN</i>	Chronic renal failure	Variable	Hyalinised glomeruli	Variable	Variable

Diabetic nephropathy

- Common cause of **chronic renal failure**
- Advanced or end stage disease occurs in up to 40% patients
- Most common lesions involve glom and are associated with three clinical syndromes:
 - Non-nephrotic proteinuria
 - Nephrotic syndrome
 - Chronic renal failure
- Diabetes also causes:
 - Hyalinising arteriolar sclerosis
 - Pyelonephritis and papillary necrosis

Cont'd

Morphology

- Capillary basement membrane thickening
- Diffuse mesangial sclerosis
- Nodular glomerulosclerosis

Pathogenesis

- Cause is due to met defect – insulin deficiency and ↑glucose
- This causes ↑ collagen type IV → thick GBM
- Nonenzymatic glycosylation of proteins

Cont'd

Tubulointerstitial changes

- Interstitial inflammation
- Papillary necrosis
- Thickening of tubular basement membrane
- Glycogen deposits in the tubular epithelial cells

TUBULAR AND TUBULOINTERSTITIAL DISEASES

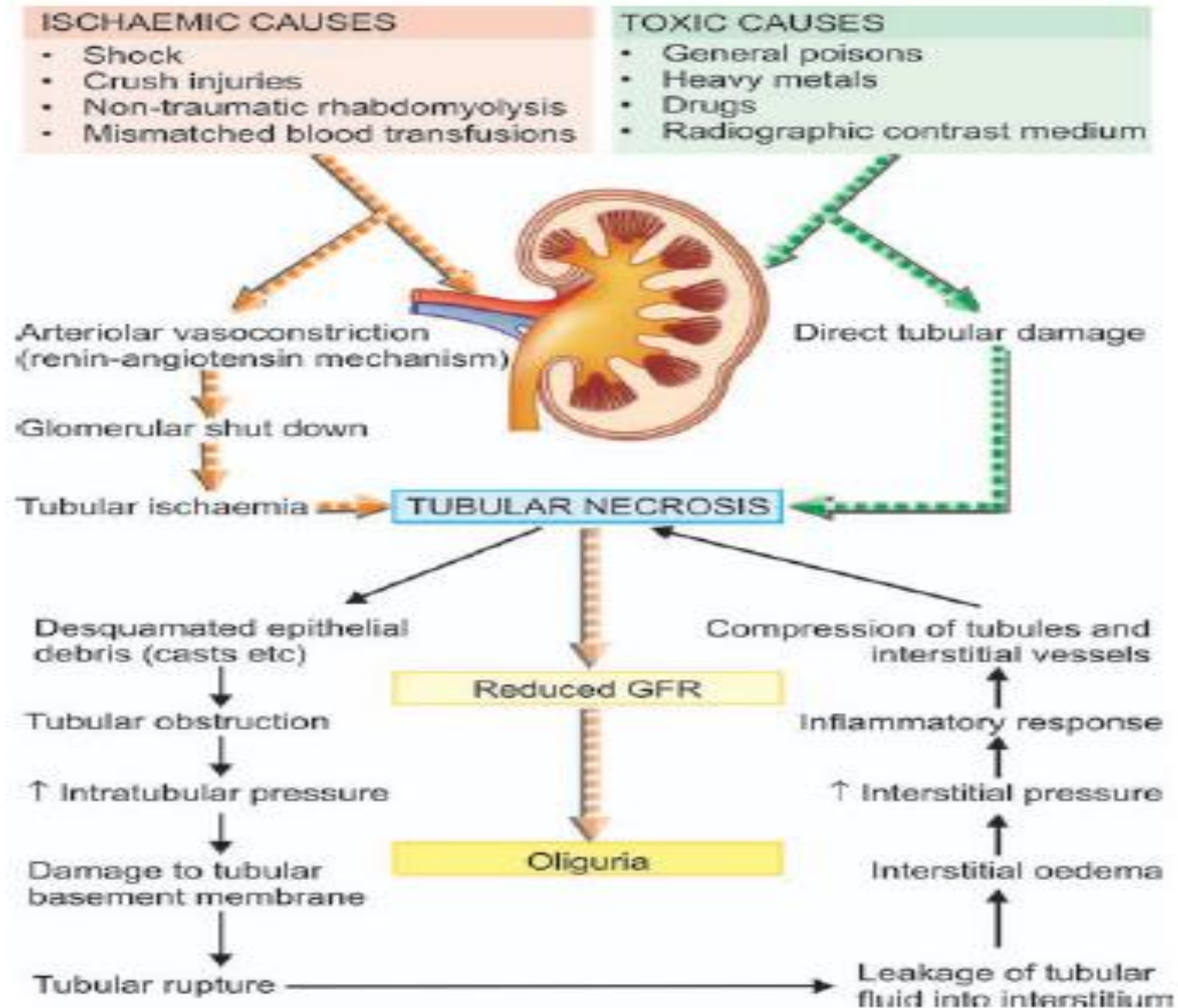
Introduction

- **I. Primary tubular diseases** that include tubular injury by ischaemic or toxic agents i.e. acute tubular necrosis.
- **II. Tubulointerstitial diseases** that include inflammatory involvement of the tubules and the interstitium i.e. tubulointerstitial nephritis.

Acute Tubular Necrosis

- **Acute tubular necrosis** (ATN) is the term used for acute renal failure (ARF) resulting from destruction of tubular epithelial cells.
- ATN is the most common and most important cause of ARF characterised by sudden cessation of renal function
- Based on etiology and morphology, two forms of ATN are distinguished — **ischaemic** and **toxic**; however both forms have a somewhat common pathogenesis.

Pathogenesis of ATN



Contrasting Features of Ischaemic and Toxic ATN

Feature	Ischaemic ATN	Toxic ATN
1. <i>Synonyms</i>	Tubulorrhectic ATN, lower (distal) nephron nephrosis, anoxic nephrosis, shock kidney	Nephrotoxic ATN, upper (proximal) nephron nephrosis, toxic ATN
2. <i>Frequency</i>	More common (80% cases)	Less common
3. <i>Major etiologies</i>	Shock, crush injuries, mismatched blood transfusion	Poisons, heavy metals, certain drugs
4. <i>G/A</i>	Kidneys enlarged, swollen, cut section cortex pale, medulla dark	Similar to ischaemic ATN
5. <i>M/E</i>	i) Distal tubular damage more prominent ii) Focal tubular necrosis iii) Regenerating epithelium iv) Casts: Hyaline, pigment, myoglobin v) Basement membrane disrupted	i) Proximal tubular damage more prominent ii) More diffuse tubular injury iii) Regenerating epithelium iv) Tubular lumina may contained dystrophic calcification v) Basement membrane generally intact
6. <i>Prognosis</i>	Worse	Good

Tubulointerstitial Diseases

- The term tubulointerstitial nephritis is used for **inflammatory process** that predominantly involves the **renal interstitial tissue** and is usually accompanied by some degree of tubular damage
- A number of primary glomerular, tubular, vascular and obstructive diseases are secondarily associated with interstitial reaction
- A number of **bacterial and non-bacterial**, **acute** and **chronic** conditions may produce tubulointerstitial nephritis

Acute Pyelonephritis

- Is an **acute suppurative inflammation** of the kidney caused by pyogenic bacteria
- Common pathogenic organism in urinary tract infection (UTI) is *Escherichia coli* (in 90% of cases), followed in decreasing frequency, by *Enterobacter*, *Klebsiella*, *Pseudomonas* and *Proteus*
- Bacteria gain entry into the urinary tract, and thence into the kidney by one of the two routes: **ascending infection** and **haematogenous infection**

Cont'd

Predisposing conditions

- Urinary tract obstruction
- Instrumentations – e.g. catheterisation
- Vesicoureteral reflux
- Pregnancy
- Gender and age – cases more common in females
- Preexisting renal lesions causing obstruction
- Diabetes mellitus
- Immunosuppression and immunodeficiency

Clinical Features

- An acute onset with chills, fever, loin pain, lumbar tenderness, dysuria and frequency of micturition
- Urine will show bacteria in excess of 100,000/ml, pus cells and pus cell casts
- Do a Urine C/S

Complications

- Papillary necrosis
- Pyonephrosis
- Perinephric abscess.

Chronic Pyelonephritis

- is a chronic tubulointerstitial disease resulting from repeated attacks of inflammation and scarring
- 1. **Reflux nephropathy**. Reflux of urine from the bladder into one or both the ureters during micturition is the major cause of chronic pyelonephritis
 - Mostly due to **Vesicoureteric reflux** is particularly common in children, especially in girls
- 2. **Obstructive pyelonephritis**. Obstruction to the outflow of urine at different levels predisposes the kidney to infection
 - Such obstruction and infection result in renal damage and scarring

Chronic Pyelonephritis

Morphology

- kidneys are usually **small and contracted** (weighing less than 100 gm) showing unequal reduction
- Generally **blunting** and **dilatation** of calyces (calyectasis) and **dilated pelvis** of the kidney

Clinical Features

- Insidious onset
- Patients may present with features of acute recurrent pyelonephritis with fever, loin pain, lumbar tenderness, dysuria, pyouria, bacteriuria and frequency of micturition
- Culture of the urine may give positive results

Renal Cyst

Classification

- Multicystic renal dysplasia
- Polycystic renal disease
 - Autosomal-dominant (adult) polycystic renal disease
 - Autosomal-recessive (childhood) polycystic renal disease
- Medullary cystic disease
 - Medullary sponge kidney
 - Nephronophthisis
- Acquired (dialysis-associated) cystic renal disease
- Localised (simple) renal cysts
- Renal cysts in hereditary malformation syndromes eg tuberous sclerosis

Polycystic Kidney Disease

Autosomal dominant (adult) polycystic renal disease

- Is a hereditary disease characterised by multiple **expanding cysts** of both kidneys that ultimately destroy the renal parenchyma and **cause renal failure**
- Familial disease, Autosomal dominant may also be due to spontaneous mutations
- Mutations in the PKD-1 gene 16p13.3 accounts for 85% cases, while PKD gene 4q21
- Cysts arise in segment of renal tubules
- End stage renal disease occurs due to pressure on parenchyma in half the patients
- Disease is universally bilateral

Genetics & Pathogenesis of PKD

- PKD1 gene encodes integral membrane protein named polycystin-1,
- function is not known but involved in cell-cell and cell-matrix interactions
- PKD2 gene product polycystin-2 is an integral membrane protein functions as a Ca^{2+} -permeable cation channel
- The mutation results in defective calcium homeostasis in the cilia of the renal epithelial tubular cells $\downarrow$ intracellular calcium ($\uparrow$ Vasopressin) & subsequent $\uparrow$ in c-AMP $\rightarrow$ IMPAIRED regulation of secretory and proliferation pathways

Cont'd

Morphology

- Bilateral enlargement of kidneys – up to 4kg
- Functioning nephrons found between cysts
- Cysts arise from tubules throughout nephron

Clinical features

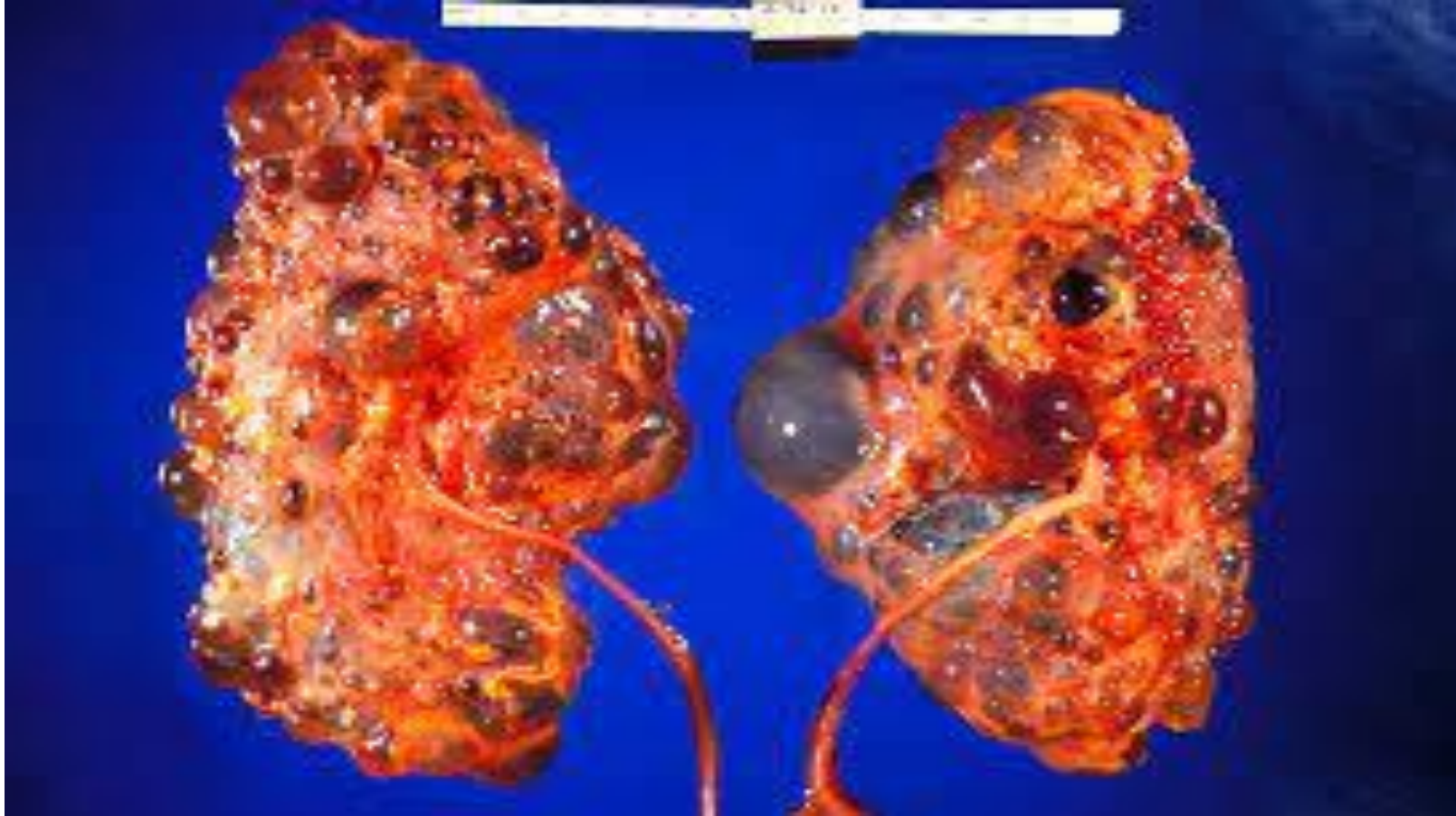
- Many patients remain **asymptomatic** until renal failure supervenes
- Pain (Flank pain) from hemorrhage or pressure effects
- Excretion of blood clots
- 40% have polycystic **liver disease**, cysts in spleen, pancreas, lungs, berry aneurysms

Cont'd

Complication

- Hypertension
- Sub arachnoid haemorrhage
- Left ventricular hypertrophy
- Mitral valve prolapse

APKD



ARPKD

- Is transmitted as an autosomal recessive trait and the family history of similar disease is usually not present
- Occurs due to a mutation in chromosome 6—6p21, PKHD1 (polycystic kidney and hepatic disease 1
- May occur in perinatal, neonatal, infantile or juvenile stage
- Perinatal and neonatal types common.
- Infant might succumb rapidly to renal failure
- Patients who survive infancy may develop hepatic fibrosis

Cont'd

Morphology

- Kidneys are enlarged
- Numerous small cysts in cortex and medulla → sponge appearance of kidney
- Dilatation of all collecting tubules
- The liver has cysts associated with portal fibrosis

Tumours of kidney

Wilms tumour (nephroblastoma)

- Common tumour of childhood.
- Peak incidence 2 - 5 years, 90% occur before 10 years of age
- 5 – 10% are bilateral
- Cure rates up to 85%

Pathogenesis and Genetics

Risk increased in groups of congenital malformations

- WAGR syndrome – aniridia, genital anomalies, mental retardation
- Denys-Drash syndrome – gonadal dysgenesis, early onset nephropathy
- Beckwith-Wiedemann syndrome – organomegaly, macroglossia, hemihypertrophy, omphalocele, adrenal cytomegaly
- Patients with WAGR and Denys-Drash syndrome carry Wilms tumour associated gene 1, WT1
- Patients with BWS have WT2

Clinical features

- Most children present with large abdominal mass
- Hematuria, abdominal pain, intestinal obstruction, hypertension
- Diagnosis – USS, CT scan, MRI, IVP, biopsy
- Treatment – surgery
 - Radiation
 - Chemotherapy
- Prognosis – overall cure 80 – 90%
- Patients under 2 years good prognosis

Renal cell carcinoma

- Tumour of adults, usually older individuals
- Also called hypernephroma, adenocarcinoma of kidney
- Majority are unilateral
- Most significant factor is tobacco smoking
- Increased risk in chronic renal failure, acquired cystic disease, tuberous sclerosis
- Most tumours sporadic, but familial ones occur:
 - Von Hippel-Lindau syndrome
 - Hereditary (familial) clear cell carcinoma
 - Hereditary papillary carcinoma

Clinical features

- Three classic diagnostic features:
 - Costovertebral pain
 - Palpable mass
 - Hematuria
- Most reliable feature is hematuria
- One of the great mimics – produces a variety of systemic symptoms unrelated to the kidney:
 - Fever, malaise, weakness
 - Paraneoplastic syndromes due to abnormal hormone production:
 - Polycythemia, hypercalcemia, hypertension, hepatic dysfunction, Cushing syndrome, eosinophilia, leukemoid reactions, amyloidosis

Cont'd

- Tends to metastasize widely before local symptoms appear
- 25% have metastases at time of diagnosis
- Metastatic sites include lungs, bone, lymph nodes, liver, adrenals, brain
- Nephrectomy is treatment of choice
- 5 year survival rate is about 45%
- Has been seen to regress in absence of treatment.
- This phenomenon also seen in gestational choriocarcinoma, melanoma, neuroblastoma

THANK YOU.